SCALE: Dependency-Aware Representation Compilation from Pretrained Transformers to Hierarchical Generative Models

A Conditional Framework with Completed Finite-Domain Mechanism Studies

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Abstract

SCALE (Size-Consistent Architecture and Latent Exchange) formulates an adaptive compiler from pretrained autoregressive Transformers to hierarchical generative models: calibrate functional representations, design a rate- and dependence-aware code hierarchy, then reuse compatible operators or perform bounded adaptation. Conditional transfer and logit-error bounds, the conditional-total-correlation limit of factorized priors, and capacity/identifiability counterexamples delimit what such a compiler can guarantee. The full pretrained, cross-size compiler remains a proposal. Completed local mechanism studies now use one small trained teacher and an exactly enumerable 16-sequence domain. Four-arm controls separated alignment from dependence, diagnosed a code-collision reconstruction ceiling, and tested balanced codec sampling and deterministic coarse-codebook maintenance. All 20 new-seed models passed fixed hard-stability and coverage gates, with minimum coarse perplexity 1.9734 of 2; the reliable teacher-alignment gate still failed. The companion JAX/Flax system records 44 passing CPU tests, resumable training, seeded generation and CPU inference measurements. These milestones validate a reproducible mechanism-study platform and a small-domain repair, not pretrained transfer, general language quality, adaptation savings or TPU performance.

A subsequent 360-model fresh-seed study across six binary source laws finds fidelity failures under every quantizer condition. On 120 frozen reference codecs, 480 training-only tabular prior fits show substantial gains from joint or sparse conditional modeling, with a capacity–estimation tradeoff. Exact invertible innovation baselines expose avoidable dependence in the benchmark; these classical constructions are not new general generative algorithms.

Keywords: cross-architecture transfer; next-concept modeling; residual vector quantization; conditional total correlation; distillation; JAX/Flax.

1 Introduction

Pretrained autoregressive Transformers contain capabilities acquired at high training cost [1, 2]. Reusing these capabilities in a different generative factorization is appealing, but tensor copying alone cannot resolve width changes, basis symmetries, incompatible operators or conditional independence assumptions. Function-preserving transformations, stitching, merging and distillation provide partial precedents rather than a universal conversion theorem [3–7].

Next-Concept Modeling (NCM) predicts multiscale discrete codes coarse-to-fine, with positions within a level generated in parallel [8]. Reconstruction can remain good while codes at the same parallel step retain strong dependence. A useful compiler must therefore co-design its representation and its generative factorization. SCALE proposes to align functional representations into a calibrated intermediate space, design a hierarchy for fidelity and predictable dependence, and transplant parameters only when an operator-level equivalence is justified.

Table 1: Notation shared with the companion manuscript. Code variables I_k are distinct from EMA counts C_{kv}. Prompt u is fixed/empty in the completed finite-domain studies.

Symbol	Meaning
B, L, W	Batch size, finest sequence length, output token vocabulary size
K, n_k, V, D	Levels, level positions, uniform codebook size, code dimension
H, J	Prior hidden width and number of Transformer blocks
I_k, i_{kbj}	Random code level and realized code ID at batch/position (b, j)
E, C, S	Codebook embeddings, EMA counts and residual sums
Q_ψ, d_ψ, p_θ	Deterministic encoder/codec map, token decoder, level prior
$p_s, p_{\theta,\psi}$	Teacher and induced student output distributions
$a, \varphi, \tau_1(a), \beta_1(a)$	Optimizer step, runtime phase, coarse temperature and commitment weight
$\mathsf{H}, \mathrm{TC}, \mathrm{TV}$	Shannon entropy, conditional total correlation, total variation
r, d_s, P_s, R_s	Canonical width, teacher width, calibration map and partial inverse; $r = D = 8$ in the pilot

This revision separates three evidence levels. The mathematical statements are conditional results. The full external pretrained compiler remains proposed. The local NCM system and a sequence of finite-domain studies have been implemented and evaluated. Those studies first exposed coverage failures, then repaired hard coarse-codebook stability, but did not establish a reliable teacher-alignment benefit. The systems companion [9] describes implementation contracts and uses the same empirical records; its results are not independent replications of this paper.

Contributions are a falsifiable compiler contract and conditional analysis; a completed paired-arm mechanism-study platform; an explicit collision-based failure diagnosis; and reproducible coverage/stability interventions with negative alignment results. No universal lossless compression, pretrained cross-size conversion or adaptation-compute reduction is claimed.

2 Related Work and Novelty Boundary

Functional correspondence. Language-model stitching and cross-size feature transplantation motivate calibration across widths [6]. CKA is a useful descriptive similarity measure, not a functional-equivalence certificate [10]. MATCHA models many-to-many layer correspondences explicitly [11]. These precedents motivate alignment; they do not establish that a single low-dimensional interface preserves a teacher's generation law.

Weight and architecture transformations. Net2Net gives function-preserving changes under specific architectures and operators [3]. Permutation-aware merging, unequal-size merging and heterogeneous fusion constrain rather than eliminate compatibility requirements [5, 12, 13]. Cross-architecture compression and Transformer-to-state-space adaptation similarly require measured adaptation [7, 14]. Distillation and restricted low-rank updates are candidate fallback mechanisms [4, 15].

Discrete parallel generation. VQ-VAE and residual codecs supply discrete representations and EMA mechanisms [16, 17]; NCM supplies the principal target factorization [8]. ResGen explores another hierarchical sampling design [18]. The conditional-TC factorization penalty below is a known information identity, also discussed by Wen et al. in the context of parallel sampling [19]. SCALE's proposed contribution is to use such constraints in an adaptive compiler and falsifiable allocation policy, not to claim that identity, NCM or model stitching as new.

3 Problem and Compiler Contract

Let teacher size s identify a model with weights $\mathcal{W}_s$, hidden width d_s, hidden functions h_s and output law $p_s(x \mid u)$ under a fixed prompt and decoding protocol. A compiler would map

$$\mathcal{C}(\mathcal{W}_s, \mathcal{A}, \mathcal{B}_{\text{res}}, \xi) \longmapsto (Q_\psi, d_\psi, p_\theta, J_s), \tag{1}$$

where $\mathcal{A}$ is calibration data, $\mathcal{B}_{\text{res}}$ is a target resource envelope, ξ fixes randomness and J_s is an audit certificate. The certificate must distinguish proved compatibility, measured approximation and untested assumptions, and account for calibration, search, codec, prior and adaptation costs. It is not implemented in full by the current experiment logs.

For fixed geometry the target code law factorizes as

$$p_\theta(i \mid u) = \prod_{k=1}^{K} \prod_{j=1}^{n_k} p_\theta(i_{k,j} \mid i_{<k}, u). \tag{2}$$

A probabilistic token decoder induces

$$p_{\theta,\psi}(x \mid u) = \sum_i p_\theta(i \mid u) d_\psi(x \mid i, u). \tag{3}$$

The local studies instead use deterministic byte argmax for d_ψ; their exact TV is specific to that decoding rule. Prompts are empty and output length is fixed. A shared representation width is a design choice, not evidence of size consistency or a lossless teacher-independent coordinate system.

4 Proposed Six-Stage Architecture and Current Implementation

1. **Audit teacher and budget.** Fix weights, tokenizer, layers, decoding, calibration splits, resource envelope and accounting rules before adaptation. The local platform records one synthetic teacher and fixed data hashes.
2. **Calibrate representations.** Seek functionally relevant maps $P_s : \mathbb{R}^{d_s} \to \mathbb{R}^r$, with optional injections $R_s : \mathbb{R}^r \to \mathbb{R}^{d_s}$. Cross-size maps may use stitching, many-to-many layers or task losses. The historical pilot uses a sign-fixed PCA map for one teacher; this is neither learned cross-teacher alignment nor a verified operator equivalence.
3. **Design the code hierarchy.** Choose (K, n_k, V, D) to balance fidelity, nominal rate, conditional TC and serial depth. The current studies fix these shapes and evaluate exact small-support dependence; adaptive architecture search is future work.
4. **Construct a level prior.** Train the shared Transformer on frozen codec labels, preserving earlier-level conditioning. This stage is implemented, as are stochastic code sampling and deterministic byte decoding.
5. **Reuse or adapt operators.** Transplant only certified compatible maps; otherwise allow restricted distillation, adapters or capacity expansion. No selective pretrained operator transplant has been demonstrated.
6. **Evaluate and allocate.** Audit hard usage, coverage, output law, quality and complete adaptation cost; reject or expand targets when gates fail. Local frozen protocols and exact diagnostics are implemented. Automated allocation, scalable density estimation and general compiler certificates remain proposed.

The distinction between a stage's executable local mechanism and its full cross-size form is essential: the local PCA/code studies do not complete SCALE.

5 Conditional Guarantees and Failure Cases

5.1 Exact transfer on an invariant compatible subspace

Proposition 1 *Let $R_s \in \mathbb{R}^{d_s \times r}$ have full column rank and $P_s R_s = I_r$. If a teacher operator F_s and canonical operator M satisfy $F_s(R_s z) = R_s M(z)$ for every reachable z, then $P_s F_s R_s = M$ on that set. If an entire stack has compatible interfaces and heads satisfying these conditions, its canonical composition is exact there.*

Proof. Apply P_s to the assumed equality and use $P_s R_s = I_r$. Repeating this identity across compatible reachable interfaces gives the composition result. □

Nonlinear attention, normalization and residual branches must individually satisfy the compatibility conditions or be included in the certified operator. A dimension match, high CKA or successful PCA reconstruction is insufficient. The present experiments did not verify these premises for a pretrained model.

5.2 A local output-error bound

Proposition 2 *Suppose $h_s = R_s z + \eta_s$, $P_s R_s = I_r$, and a reconstructed canonical vector obeys $\|\widehat{z} - P_s h_s\| \le \epsilon_Q$. Let a compatible logit map g be L_g-Lipschitz into the infinity norm, with teacher logit error at most ϵ_T relative to $g(z)$ and student logit error at most ϵ_D relative to $g(\widehat{z})$. Then their logit vectors satisfy*

$$\|\ell_s - \ell_{\theta,\psi}\|_\infty \le \Delta = \epsilon_T + L_g(\|P_s\|\,\|\eta_s\| + \epsilon_Q) + \epsilon_D, \tag{4}$$

and the KL between their softmax laws is at most 2Δ.

Proof. Since $P_s h_s = z + P_s \eta_s$, the canonical error is at most $\epsilon_Q + \|P_s\|\|\eta_s\|$. Apply Lipschitz continuity and the triangle inequality. Log-sum-exp is 1-Lipschitz in the infinity norm, so each log-ratio of the two softmax laws is at most 2Δ; average under the teacher law. □

This is a one-interface bound with common conditioning and a compatible logit head. It is not a certified free-running sequence bound. A sequential chain rule would require uniform conditional control on all relevant teacher prefixes; a hierarchical mixture needs its own coupling or output-law argument. None of these constants has been certified for the local or proposed pretrained models.

5.3 Conditional total correlation limits parallel factorization

Proposition 3 *For a fixed discrete law of $(I_{<k}, I_k)$, the best unrestricted within-level factorized conditional approximation satisfies*

$$\min_{\{q_j\}} \mathbb{E}_{I_{<k}} D_{\mathrm{KL}}\left(P(I_k \mid I_{<k}) \middle\| \prod_j q_j(I_{k,j} \mid I_{<k}) \right) = \mathrm{TC}(I_k \mid I_{<k}). \tag{5}$$

The minimizers are the true conditional marginals.

Proof. Expanding the expected log ratio gives conditional TC plus $\sum_j \mathbb{E}_{I_{<k}} D_{\mathrm{KL}}(P(I_{k,j} \mid I_{<k}) \| q_j)$. Each extra KL is nonnegative and vanishes at its conditional marginal. □

For a single shared unbiased bit copied into two positions, the law puts mass one half on each of 00 and 11. Its best parallel product has a $\log 2$-nat KL gap even though the pair is perfectly reconstructable. Neural parameter sharing and finite capacity can add error above this lower bound. Reducing TC by collapsing codes can destroy information, so TC must be evaluated together with reconstruction, support and hard usage [19].

5.4 Finite capacity and empirical collision ceilings

Proposition 4 *If a deterministic codec has at most M codewords and must exactly reconstruct $N > M$ distinct inputs through a deterministic decoder, perfect reconstruction is impossible. For any fixed codec and input law p_s, the best deterministic sequence reconstruction probability is*

$$\max_d \Pr[d(Q_\psi(X)) = X] = \sum_i \max_{x:Q_\psi(x)=i} p_s(x), \tag{6}$$

with empty cells contributing zero.

Proof. Pigeonhole requires a collision when $N > M$. In each code cell the decoder can select at most one input as its exact reconstruction. Selecting the most likely member of every nonempty cell attains the sum; no other choice can exceed it. □

The nominal capacity $\sum_k n_k \log_2 V$ bits bounds code cardinality; it is not an entropy-coded byte size and excludes fallback and metadata. Conversely, enough nominal codewords does not prevent learned collisions. Equation (6) explained the completed decoder-replay failure below without relying on a population capacity shortage.

5.5 Calibration does not identify arbitrary off-support behavior

Lemma 1 *For a finite calibration set $\mathcal{A}$ and unrestricted function class, agreement on $\mathcal{A}$ does not determine behavior at an unobserved point $x^* \notin \mathcal{A}$.*

Proof. Choose two functions equal on every calibration point but different at x^*. Their calibration observations coincide, so an observer restricted to those observations cannot distinguish them. □

This limits guarantees from black-box calibration observations, not all algorithms with full access to teacher weights. Structural assumptions or broader tests may supply information beyond that finite sample. The local studies enumerate their entire small domain, but do not test new domains.

6 Training Objectives and Hard-Usage Safeguards

A proposed compiler objective combines masked reconstruction, VQ commitment, representation alignment, dependence, marginal entropy and a resource penalty:

$$\mathcal{L}_{\text{codec}} = \mathcal{L}_{\text{rec}} + \beta\mathcal{L}_{\text{commit}} + \lambda_A\mathcal{L}_{\text{align}} + \lambda_D\mathcal{L}_{\text{TC}} - \lambda_H\mathsf{H}(q_{\text{marginal}}) + \lambda_R\mathcal{R}. \tag{7}$$

Here $\mathcal{R}$ is a proposed cost/rate term; no adaptive resource search was run. The local implementation has explicit loss coefficients, masked neural and EMA updates and fixed geometry. The four arms change alignment/dependence coefficients while executing a common graph. The frozen prior minimizes $-\log p_\theta(Q_\psi(x))$. Optional decoder-only replay changes neither codec assignments nor prior parameters.

Marginal entropy and conditional TC have different roles: a constant code can have zero TC and terrible fidelity; independent uniform codes may be diverse but uninformative. Soft assignment entropy can improve while hard usage remains collapsed. The completed maintained coarse path therefore tracks hard counts independently, fits initial centroids by deterministic K-means++ and uses current-batch latent replacements [20]. It schedules $\tau_1(a)$ from 2 to 0.25 and $\beta_1(a)$ from 0 to 0.25 over 200 steps, and adds a coarse soft marginal-entropy coefficient of 0.05. Quantization stays hard in the forward pass; there is no Gumbel-Softmax relaxation.

The separate coarse centroid group applies $E_1' = E_1 + \eta_{\text{coarse}}(E_1^{\text{EMA}} - E_1)$, with $\eta_{\text{coarse}} = 0.25$. This scale is relative to the EMA centroid proposal, not to the neural Adam learning rate.

Table 2: Synchronized single-device CPU median generation latency (ms), batch size 16, float32, five warmups and 30 repetitions. NCM includes byte decoding; AR uses a KV cache. Length 8 uses trained toy models of width 16 and one block; lengths 128/512 use untrained width-64, two-block shape checks. These are not quality-matched speedups or TPU measurements.

Bytes	Scope	Padded NCM	Actual-length NCM	Cached AR
8	Trained toy	0.489	0.403	0.937
128	Untrained shape	26.877	14.390	39.588
512	Untrained shape	173.887	85.831	299.951

Fine EMA and neural rates are unchanged. Initialization flags, serializable random keys, hard-usage/replacement buffers, optimizer state and schedule counters are checkpointed. All arms invoke the same masked maintenance graph, and exact same-backend resume remains tested. The systems companion gives the detailed state contract [9].

7 Completed CPU Experiments and Finite-Domain Results

7.1 Evidence snapshot and completed engineering milestones

Historical results before the research-extension subsections describe the implementation at revision `1d600d7`, dated November 12, 2026, package version 0.1.0 [21]. The later extensions have separately archived protocols and source hashes. The historical recorded stack was Python 3.12.16, JAX/jaxlib 0.10.2, Flax 0.12.8, Optax 0.2.8 and NumPy 2.5.4, on macOS 26.6.2 arm64 CPU, with float32 arrays. The CPU model was not recorded; no TPU hardware was available. The validation record reports 44 passing pytest cases, including parameterized runs, exact checkpoint continuation and two virtual four-device CPU sharding checks. Test runtime is not an algorithmic benchmark.

The completed engineering path includes UTF-8 byte loading and chunking, configuration/-data hashes, atomic checkpoint replacement, step-addressed random batches, a frozen-codec prior, seeded categorical code sampling with temperature and optional top-k, and byte decoding. Output length remains a caller input; there is no prompt-conditioning, learned-length or EOS interface. The separate 26-byte training and 13-byte validation fixture reached 100% training-byte reconstruction by step 100 of a 500-step run. Validation CE increased from 1.949 nats at step 100 to 3.550 at step 500: this was overfitting, not a successful held-out language result. The historical codec/prior logs are now archived in `reports/text-experiments/`; their checkpoint hashes match the CPU benchmark's source identities. Those records are distinct from the finite-domain teacher studies below.

7.2 CPU inference measurements

The benchmark timed lowered/compiled functions separately from warmed inference, kept timed inputs on-device and waited for every output with `jax.block_until_ready`. Sampling used temperature 1.0; host text conversion and checkpoint I/O were excluded. NCM byte decoding and all cached AR serial steps were included. With batch 16 and untrained length-512 models, actual-level execution reduced median latency from 173.9 to 85.8 ms. The trained length-8 fixture stored 21,328 codec parameters, 4,032 prior parameters and 768 codebook embedding elements, whereas AR had 11,904 parameters. Neither parameter count nor quality was matched. Longer shape checks used $D = 32, V = 64, H = 64, J = 2$, four attention heads and levels $(L/4, L/2, L)$; they were not trained on text.

Compiler live-buffer estimates, process-lifetime counters where supported, compilation times and 18 separate profiling traces were recorded. Per-case measured peak device memory was

Table 3: Completed finite-domain protocols. Known seeds are repeated initializations, not new independent confirmations. Every four-arm follow-up fits 32 student models, of which 20 use newly reserved student seeds; the teacher, domain and sampled datasets remain fixed.

Protocol	Known-seed checks	Evaluation seeds	Models
Initial pilot	–	0,1,2	9
Decoder replay	0,1,2	3,4,5,6,7	32
Balanced codec	0,4,7	8,9,10,11,12	32
Coarse maintenance	4,8,10	13,14,15,16,17	32

unavailable; compiler estimates are not peak memory measurements. These observations establish a local implementation improvement, not TPU acceleration or a quality-matched model speedup.

7.3 One-teacher finite-domain protocol

Each domain input is a length-4 sequence over A/B (16 possibilities). A locally trained causal byte Transformer has width 16, two heads and one block; it is not an external pretrained language model. Its source distribution mixed probability $1-\rho$ equally over AAAA/BBBB with $\rho = 0.25$ uniformly over all 16 sequences. After 400 teacher updates, its A/B logits were normalized at every prefix to define an exactly enumerable law p_s. Teacher seed 2026 and its checkpoint remained fixed in every protocol.

Training, validation and calibration comprised 512, 512 and 256 independently sampled sequence draws, respectively. They shared the finite support and could repeat identical sequences across splits; there was no held-out domain or larger-teacher evaluation. A sign-fixed PCA interface fitted on calibration features mapped normalized causal teacher features to $r = D = 8$. Student codebooks had $K = 2$, $(n_1, n_2) = (1, 4)$, $V = 2$, and nominal capacity 5 bits; student priors used width 16, two heads and one block. The byte vocabulary was $W = 256$, with no fallback side channel in these studies. The AR teacher used BOS ID 256 on its input only.

Arms were R (reconstruction/commitment/utilization), A (R plus representation alignment), D (R plus dependence) and AD (both). Paired initial states, minibatch order, architecture and update counts were fixed within each seed. Every student used 400 codec updates and 400 frozen-codec prior updates, global batch 64 and Adam learning rate 0.003. The three four-arm follow-ups additionally used 200 decoder-only replay updates at the same rate. Prior parameters and code assignments stayed fixed during replay. Shared teacher/calibration preparation was charged equally; zero auxiliary coefficients still executed the common student objective graph. Equality of compiler-estimated student step FLOPs is not equality of measured total hardware work or a compute-saving claim.

7.4 Metrics and hard-code diagnostics

For a teacher output law p_s and induced student law $p_{\theta,\psi}$, the reported output error is

$$\mathrm{TV}(p_s, p_{\theta,\psi}) = \tfrac{1}{2} \sum_x |p_s(x) - p_{\theta,\psi}(x)|. \tag{8}$$

Every codeword is enumerated and passed through the deterministic byte-argmax decoder. An extra output bin retains probability outside the A/B domain. Missing teacher support is $\sum_{x:p_{\theta,\psi}(x)=0} p_s(x)$. If this is positive, forward KL is infinite; JSON reports a null KL together with the missing mass, not a smoothed finite substitute.

Table 4: Mean exact output total variation (TV; lower is better). Columns R, A, D, and AD denote reconstruction, alignment, dependence, and their combination. Only the listed student seeds enter each row; the first row has no decoder replay, later rows include 200 replay updates. Different rows use different seed sets and interventions, so row-to-row means are not causal quality comparisons.

Study	Student seeds	R	A	D	AD
Initial pilot	0,1,2	0.3316	0.3339	–	0.1744
Decoder replay	3,4,5,6,7	0.2417	0.1453	0.2472	0.1577
Balanced codec	8,9,10,11,12	0.4257	0.2370	0.5229	0.2468
Coarse maintenance	13,14,15,16,17	0.1381	0.1318	0.1392	0.1342

Hard conditional dependence is evaluated under the teacher-induced code law $P_{s,\psi}(i) = \sum_{x:Q_\psi(x)=i} p_s(x)$:

$$\mathrm{TC}(I_k \mid I_{<k}) = \sum_{j=1}^{n_k} \mathsf{H}(I_{k,j} \mid I_{<k}) - \mathsf{H}(I_k \mid I_{<k}). \tag{9}$$

Exact entropies are checked against an independent factorized-prior KL calculation. The training surrogate instead enumerates a mixture of soft assignment products; its gradients and numerical value do not certify the hard code law. Enumeration is capped at 4096 codewords.

Byte reconstruction averages correct valid positions; sequence reconstruction requires every valid byte to match. These are separate metrics. Coarse perplexity is $\exp \mathsf{H}(p_v^{\mathrm{hard}})$, using hard assignments; the theoretical maximum is V. Population values weight all domain inputs by p_s; monitored minibatch counts use actual global training assignments. Teacher-forced code CE conditions on true earlier codes, whereas free-prefix CE uses greedy earlier predictions. Their difference diagnoses exposure error; neither is directly comparable to an AR next-byte perplexity, and paired free-running byte accuracy is not a general quality score for unconditional samples.

7.5 Completed controls, negative results and coverage repair

The initial three-seed pilot passed its fixed AD-versus-A TV/dependence/fidelity gates. Mean hard conditional TC fell from 0.4550 to 0.1368 nats. That comparison lacked a D-only control; most TV gain came from one seed, and seed 0's AD model missed 5.12% of teacher probability. It therefore did not establish a reliable alignment contribution or full SCALE.

The four-arm decoder-replay protocol reserved student seeds 3–7. Its coverage and reliable-alignment gates both failed. In AD seed 4, 16 inputs occupied only 10 codewords. The deterministic code-to-sequence reconstruction ceiling $\sum_i \max_{x:Q_\psi(x)=i} p_s(x)$ was 89.80%; replay achieved 87.87% sequence reconstruction and still missed 6.63% of teacher mass. More decoder updates could not recover distinctions already merged into the same code.

The balanced-codec protocol reserved seeds 8–12. Codec reconstruction and alignment sampled uniformly over distinct observed training sequences; self-normalized training-frequency importance weights kept soft dependence aimed at empirical teacher mass. Prior training retained the original empirical frequency distribution. No population probabilities or validation examples selected repair targets. All five new AD runs retained 16 encodings and achieved 100% sequence reconstruction and complete output support, already before extra replay. The known failing seed 4 also recovered under the changed sampler. The reliable-alignment gate nevertheless failed: AD beat D on only 3/5 new seeds, and A had slightly lower mean TV than AD. Thus the earlier dependence benefit was not uniform across samplers and initializations.

Table 5: Matched historical-seed ablations. Reconstruction is sequence accuracy in percent; usage failures exceed 5 percent collapsed minibatches. All objectives and schedules remain shared.

Condition	Recon.	Usage failures	Oracle TV
Minimal EMA	100.000	7/20	0.2685
EMA + recovery	99.953	0/20	0.1229
Reference	99.953	0/20	0.1215
Fixed binary	100.000	0/20	0.1222

7.6 Coarse-codebook stabilization

The maintained protocol reserved seeds 13–17; seeds 4, 8 and 10 were known-case checks. It combined two initial encoder batches, 20 K-means++/Lloyd iterations, independent hard-usage EMA decay 0.9, threshold 1 assignment per global batch, 25-step dead-code grace, reset count 2 and replacement noise 0.01. It added coarse entropy coefficient 0.05, temperature 2.0 to 0.25 and commitment 0 to 0.25 over 200 steps, and centroid step scale 0.25. Fine-book EMA and neural Adam rates were unchanged. All four arms executed the same maintenance logic. The individual components were not separately ablated.

The fixed stability gate required both codes active, population coarse perplexity at least 1.5, no more than 5% collapsed minibatches after the warmups, complete teacher output support and sequence reconstruction at least 98%. All 20 new-seed models (five seeds times four arms) passed. Minimum population perplexity was 1.9734 out of 2; each model recorded zero collapsed steps over 200 post-annealing observations. Both codes remained active and output support was preserved. AD minimum sequence reconstruction was 99.07%; the other three arms reached 100% on the new seeds. All 32 runs, including known-case checks, had matching compiler-estimated per-step FLOPs: 10,564,359 for codec training, 6,996,102 for prior training and 5,837,030 for decoder replay. Initialization used 128 valid pooled samples per model and was shared per seed, with its budget charged equally.

The reliable teacher-alignment gate was distinct: AD had to beat D by mean TV at least 0.02 and win on at least 4/5 new seeds, while preserving coverage. It won 4/5 but its mean improvement was only 0.00498, so that gate still failed. Stable codebooks are an engineering achievement, not evidence of cross-size transfer, reduced total adaptation compute or general language-model quality. All gates were saved locally before their study; no external preregistration is claimed.

7.7 Matched maintenance ablations and fresh source distributions

The historical 20-model stability result above remains conditional on its original domain. A subsequent 100-run paired ablation reused seeds 13–17, 400 codec updates and 200 decoder replay updates per model. It compared ordinary EMA without recovery, ordinary EMA with recovery, the original slowed-centroid reference, and fixed coarse binary endpoints. Every condition retained the same data-driven initialization, entropy objective, temperature and commitment schedules, neural optimizer and four objective arms. Thus it isolates centroid movement/recovery, not the necessity of all regularizers. Unguarded EMA reconstructed perfectly but failed the usage-window criterion in 7/20 runs. Recovery and fixed endpoints were promising within that envelope; small CPU timing differences did not establish substantial acceleration.

A separately specified follow-up used fresh student seeds 31–35, six full-support binary laws and lengths four or eight: the original constant-mode source, independent Bernoulli(0.2) symbols, a persistent Markov chain, noisy parity, skewed constant modes and alternating modes. All three stable quantizers were trained separately on each source with the same 512/512/256 training, validation and calibration draw budgets. Six independently initialized small teachers supplied

Table 6: Fresh seeds 31–35 across all four objective arms. Cells give mean factorized-oracle output TV and fidelity failures out of 20 models in parentheses. No prior is trained in this study.

Source	EMA + recovery	Reference	Fixed binary
Original source	0.143 (2)	0.127 (1)	0.119 (0)
Biased independent	0.087 (1)	0.093 (1)	0.089 (0)
Markov	0.485 (4)	0.489 (6)	0.491 (0)
Noisy parity	0.470 (4)	0.470 (4)	0.493 (4)
Skewed modes	0.182 (2)	0.181 (2)	0.178 (3)
Alternating modes	0.705 (3)	0.740 (3)	0.706 (4)

Table 7: Training-only prior fits on frozen reference codecs passing the sampled fidelity check. Lower decoded output TV is better. All-codec results, including flagged failures, remain in the public report.

Source	Eligible	Indep.	Tree	Joint
Original source	19	0.1359	0.0771	0.0465
Biased independent	19	0.0955	0.0869	0.0527
Markov	15	0.4956	0.1232	0.1444
Noisy parity	16	0.4647	0.4570	0.2308
Skewed modes	18	0.1746	0.1128	0.0840
Alternating modes	17	0.7526	0.2596	0.1416

alignment features; the analytic source law, rather than the fitted teacher law, supplied training samples and population evaluation. Consequently these TV values are not direct before/after comparisons with the historical teacher-law pilot. All 360 models received 400 codec and 200 replay updates. The study is broader-distribution mechanism validation, not natural-language evaluation or transfer to an unseen domain. No condition passed every fidelity check. The fixed binary condition incurred 11/120 model failures, EMA with recovery 16/120 and reference 17/120; those counts do not imply a uniform winner, since failure severity differed. In noisy-parity seed 31, AD fixed endpoints reconstructed 25% of source mass and missed 75% under the factorized oracle. The checks require sequence accuracy at least 98%, missing/outside mass at most 1%, and collision-based fidelity ceiling at least 98%. Balanced coarse usage is diagnostic, not a universal criterion for skewed sources. Production defaults remain unchanged.

7.8 Empirical dependency models on frozen reference codecs

The next experiment fitted four tabular priors to the original 512 training code observations of each of the 120 reference codecs: independent fine codes, adjacent pairs, a conditional dependence tree and a joint fine-level table. No neural or codec update was performed. Every conditional row receives total symmetric Dirichlet pseudo-count one. Tree topology uses empirical conditional mutual information from training codes only and is rooted at a center. This is related to Chow–Liu approximation [22]; the shared topology and smoothed tables are not asserted to retain its unrestricted optimality guarantee.

A sampled fidelity filter requires validation sequence accuracy and training collision ceiling at least 98%. It flags 16 codecs and accepts 104, including one subsequently found below the population reconstruction threshold. The filter is not a population certificate. All 480 prior fits and all-codec results remain archived; population probabilities are used only for evaluation, not fitting, topology, smoothing selection or the eligibility filter. Joint tables improve mean decoded TV over independent tables on every source. On Markov data, the conditional tree instead beats the joint table, consistent with a beneficial sparsity/estimation tradeoff. At length eight

the independent, pair, tree and joint families have 17, 25, 31 and 511 free parameters including the coarse distribution. Independent, pair and joint sampling take two rounds; observed center-rooted trees take three to six. Enumeration/fit timings are not end-to-end generation latency. Full joint tables grow exponentially and remain finite-domain controls, not a proposed general language architecture.

7.9 Exact innovation coordinates and the alignment-target diagnosis

For the tested symmetric binary Markov law, define

$$y_1 = x_1, \qquad y_t = x_t \oplus x_{t-1} \quad (t > 1). \tag{10}$$

The variables are independent, with probabilities 0.5 for the initial bit and 0.15 for subsequent flip indicators. Prefix XOR reconstructs every input. For the noisy-parity law (90% uniform even parity plus 10% uniform), let

$$y_i = x_i \ (i < L), \qquad y_L = \bigoplus_{i=1}^{L} x_i. \tag{11}$$

The first variables are independent fair bits and the last is an independent Bernoulli(0.05) error bit. Both are bijections; exhaustive enumeration verifies exact reconstruction and factorization to floating-point roundoff. All latent bits can be drawn independently, followed by deterministic decoding. Prefix XOR admits linear work and logarithmic parallel depth via associative scan [23]. These are classical known-structure constructions, not newly learned transforms or universal independence theorems.

For any finite bijection T, distributional KL is invariant:

$$D_{\mathrm{KL}}\Big(P_X \Big\| (T^{-1})_{\#} q\Big) = D_{\mathrm{KL}}(P_{T(X)} \| q). \tag{12}$$

This removes codec collision error and makes representation choice directly relevant to factorization. Arbitrary finite distributions need not admit independent fixed-length coordinates under a deterministic bijection, and an inverse can itself be costly. These simple sources therefore motivate information-preserving baselines without establishing a solution for language.

A separate replayable diagnostic examines the historical alignment target. The teacher sees shifted causal inputs, whereas the codec sees the full text. Its complete projected target is identical for sequences differing only in the final byte, although that byte retains 0.4293 conditional bits. PCA/head reconstruction changes teacher predictive probabilities by only 2.33×10^{-6} mean KL nats; alignment substantially improves feature matching but cannot make this target information-complete. This identifies a conditioning conflict, not a controlled causal explanation of all failures.

7.10 Evidence scope, reproducibility and next steps

The 480 fits reuse 120 codecs; objective arms share seeds and datasets. They are not 480 independent replications. Paired analysis should treat student seeds as replication units. The broader studies adapt separately to each small binary source and retain the historical alignment convention and shared regularizers. No pretrained transfer, language quality or TPU result follows. Reconstruction constraints and uncertainty-aware acceptance should precede quantizer replacement; sparse dependencies and small joint blocks are the next tractable modeling controls. The current extension suite passes 62 tests; the historical 44-test record above is retained separately. Compact frozen codec tables reproduce all 480 fits without ignored checkpoints; compressed feature arrays reproduce the alignment diagnosis. Protocols, raw observations, conditional structures, source/checkpoint hashes, source-law controls and scripts are archived

under `reports/maintenance-ablation/`, `reports/fresh-domain-validation/`, `reports/dependency-validation/`, `reports/alignment-diagnosis/` and `reports/invertible-baselines/`. One float32 entropy-identity verification stopped after 268 completed models; float64 verification agreed with direct KL to 1.11×10^{-15}. Training helpers and precision were unchanged; saved results and the affected checkpoint were resumed, with no seed replacement. One timing is missing and retained as null. The interruption and continuation are part of the evidence record.

8 Falsifiable Next-Stage Evaluation

Reliable contribution of alignment. The immediate scientific question is why AD has not reliably improved over D. Use new paired student seeds under one fixed teacher/data/sampler protocol; inspect representation targets and objective scales, and test alignment without changing the coverage/stability package. Do not select successful seeds after observing the gate. Decompose the maintenance package with compute-accounted ablations before attributing the repair to any one component.

External pretrained transfer. Only after a held-out real-corpus codec baseline is stable, evaluate multiple open-weight teacher sizes (for example a Pythia family) and held-out domains [2]. Compare scratch NCM, teacher-logit distillation, feature-only alignment, dependence-only adaptation, copied-but-unverified operators, certified reuse where possible, and SCALE's full policy. Teacher sizes, tokenizer rules, calibration quantities and adaptation limits must be fixed before each study. No such pretrained-size comparison has been completed.

Dependency and capacity. Vary serial depth, codebook size and canonical width at matched parameter and nominal-rate budgets. Measure hard conditional TC, reconstruction, collisions, output support, teacher-forced/free-prefix errors and decoded output quality. The exact small-domain surrogate enumerates at most 4096 codewords and cannot be assumed scalable. Learned density critics or other estimators require independent checks of bias, variance and optimization behavior.

Cost and failure criteria. Record teacher preparation, feature extraction, calibration, architecture search, all failed adaptation attempts, compilation, updates and I/O. A reused teacher's sunk pretraining cost should be stated separately from additional adaptation cost and from a scratch baseline. Reject claims of savings if matched-quality adaptation cost is not lower. Reject universal canonical alignment if measured functional residuals or required canonical width grow beyond the resource budget. If no fixed-budget hierarchy preserves support and useful free-running quality, report the failed envelope rather than silently increasing it.

Hardware. Carry synchronized CPU measurements to actual TPU hardware before claiming TPU throughput, memory efficiency or multi-host scaling. Match quality, precision, batch, output lengths, parameter budgets and decoding rules. Report compilation and steady-state latency separately, with uncertainty, peak memory, collectives and input/checkpoint overhead. Virtual CPU meshes are correctness checks.

9 Discussion, Threats to Validity and Future Work

The completed work moves SCALE from an exclusively conceptual framework to a small, auditable mechanism-study platform. The collision analysis distinguishes recoverable information

at the input from distinctions retained by the codec. Balanced sampling restored coverage; maintenance stabilized both hard coarse codes. Neither result proves that aligned teacher features improve a parallel prior, and the final negative alignment gate must guide the next experiment.

The historical studies have one teacher, one finite support, fixed datasets and changing student seed cohorts across follow-ups. Split names denote independent draws, not unseen sequences. Paired arms support within-protocol contrasts; differing cohorts, samplers and maintenance settings confound cross-protocol mean changes. The 20-model stability result is conditional on this domain and the combined package. Compiler-estimated step FLOPs and local gates are not measured total compute or externally preregistered confirmation. The text fixture overfit.

The conditional propositions describe assumptions that a real compiler must verify; they are not certificates supplied by PCA. Different widths may lack a common invariant functional subspace. Tokenizers, attention patterns and normalization can defeat exact transplantation. Hierarchical sampling can amplify local errors, and hard quantization can remove low-probability states. The present deterministic output decoder is a documented evaluation protocol, not a calibrated general language distribution.

Future work includes real-corpus held-out tests, further regularization ablations, robust alignment controls, scalable dependence estimation, cross-size learned interfaces, selective operator transplantation and complete adaptation accounting. Prompt/length/EOS interfaces, serialized mixed byte routing, Rust orchestration, TPU profiling and multi-host recovery remain systems work. A useful SCALE result must demonstrate fidelity and cost together under declared gates.

10 Conclusion

SCALE remains a conditional, falsifiable proposal for cross-size representation compilation. Completed CPU engineering and finite-domain controls now supply a reproducible testbed, collision diagnosis and a coarse-codebook stabilization result across 20 new-seed models. The reliable alignment gate still failed despite stable usage and support. These outcomes refine the next experiments without establishing pretrained transfer, language-model quality, compute savings or TPU performance.

The newer source-law studies qualify the original stability envelope: no quantizer passes all fidelity checks. Empirical dependency priors improve output distributions without changing the codec, while exact innovation coordinates demonstrate that some benchmark dependence is representation-induced. The current priority is information preservation and dependency design, with production defaults unchanged, rather than quantizer replacement. These studies remain finite-domain controls, not language or hardware validation.

A Availability, Reproduction and Authorship

Implementation revision `1d600d7`, package version 0.1.0, is the historical evidence baseline; source manifests identify the current experimental extensions [21]. The systems companion [9] and this paper share `shared_notation.tex`, `shared_metrics.tex`, `shared_experiments.tex` and report-derived tables. They describe the same experiments rather than independent replications. The original SCALE artifact was PDF-only; an earlier revision supplied editable LaTeX. The present revision extends its evidence while retaining the conditional analysis and proposed compiler scope.

`paper/README.md` documents both builds and the cross-paper consistency audit. `paper/generate_results.py` reads committed evidence, while `paper/references.json` is the common bibliographic registry. Bibliographies are generated only from cited entries; no BibTeX step is needed. The implementation and manuscript manifests identify inputs independently

of the mutable branch. There is no archival DOI. Fabian Balensiefer is the author; underlying NCM, VQ and transfer methods are credited to their cited authors.

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